Architecture Diagram – Client-Server Three-Tier Architecture

React · Django REST Framework · PostgreSQL

Tier 1 – Presentation (Client)

Web browser

Web browser → React Single-Page Application (SPA)

- **Citizen UI** — Registration, requests, tracking
- **Employee UI** — Department request processing
- **Department Manager UI** — Services, employees, KPIs
- **System Administrator UI** — Departments, managers, settings

React · React Router · Axios · MUI (RTL) · Stripe.js

HTTPS · REST/JSON · multipart (file upload)

Tier 2 – Application (Server)

Modular Monolith

Django + Django REST Framework

API Layer – REST/JSON

URL Routing · ViewSets / APIViews · Serializers · Permissions · Validation

Business Logic Layer – Django Apps

- **accounts** — Accounts & roles
- **departments** — Departments
- **services** — Service builder
- **requests** — Request life cycle
- **archive** — Archiving
- **payments** — Payment gateway
- **notifications** — Email alerts
- **feedback** — Ratings, complaints
- **reports** — KPIs & reports
- **audit** — Activity log

Data Access Layer – Django ORM · Storage API

Security & Permissions

- Session authentication (HttpOnly cookie)
- CSRF protection
- Role-based access control (RBAC)
- Department data isolation on server
- Password hashing
- Operation logging

External Services

Stripe

Payment gateway (Test Mode)

Checkout page

Stripe API (create checkout session)

Signed Webhook (payment result)

SMTP

SMTP Server

Mail server (notifications only)

File read / write

SQL via Django ORM (psycopg)

Tier 3 – Data

PostgreSQL · server file system

Media File Storage

ID card images and request attachments (server file system)

PostgreSQL

Accounts, departments, services, requests, payments, logs